

LAB 01 Solution Report

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name	TANEES ZAHRA
Reg No	2024-BSE-55
Course Code	
Section	A
GitHub Username	taneeszahra

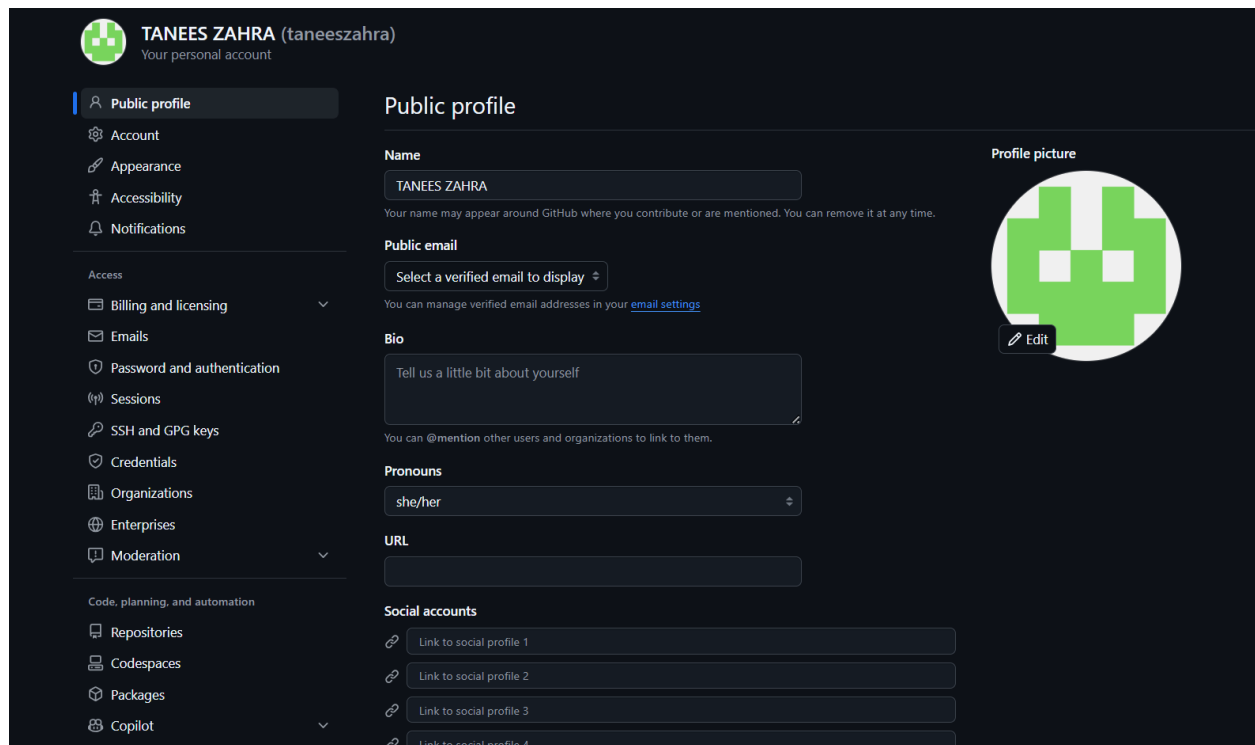
Task 1: Virtual Machine Creation

- **GitHub Profile**

GitHub profile page displaying the professional username and account details.

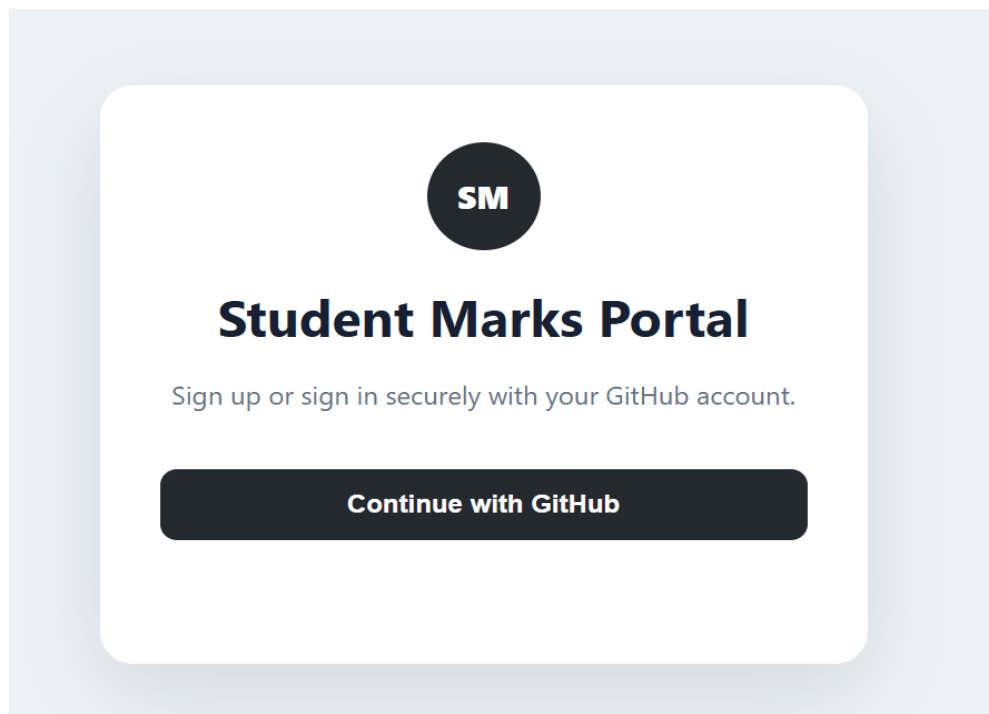
The screenshot shows the GitHub profile page for the user 'taneeszahra'. The page has a dark theme. At the top, there's a navigation bar with the user's name and a search bar. Below the navigation bar, there's a section for 'Popular repositories' which states 'You don't have any public repositories yet.' To the left of this section is a large circular profile picture featuring a green and white pixelated GitHub logo. Below the profile picture, the user's name 'TANEES ZAHRA' is displayed, followed by their GitHub handle 'taneeszahra · she/her' and an 'Edit profile' button. To the right of the profile picture, there's a section titled '2 contributions in the last year' with a calendar grid showing contributions for the months of September, October, November, and December. The calendar grid shows a green square for a contribution on Wednesday, December 1st, and another green square for a contribution on Friday, August 1st. Below the calendar grid, there's a section titled 'Contribution activity' for 'September 2026', which states 'taneeszahra has no activity yet for this period.' and a 'Show more activity' button. At the bottom of the page, there's a footer with the GitHub logo, copyright information, and links to Terms, Privacy, Security, Status, Community, Docs, Contact, Manage cookies, and Do not share my personal information.

GitHub public profile settings showing the completed actual name field.



Task 2: Register in the Student Marks Portal

Student marks portal displaying the connected GitHub profile information and account details.



Student portal showing the successfully submitted course enrollment status awaiting approval.

STUDENT MARKS PORTAL

Academic Record

Sign out



TANEES ZAHRA

GitHub authenticated

Registration

2024-BSE-55

GitHub username

taneeszahra

Email

taneeszahra07@gmail.com

Approved courses

1

+ Enroll in another course

APPROVED COURSES

Course Marks

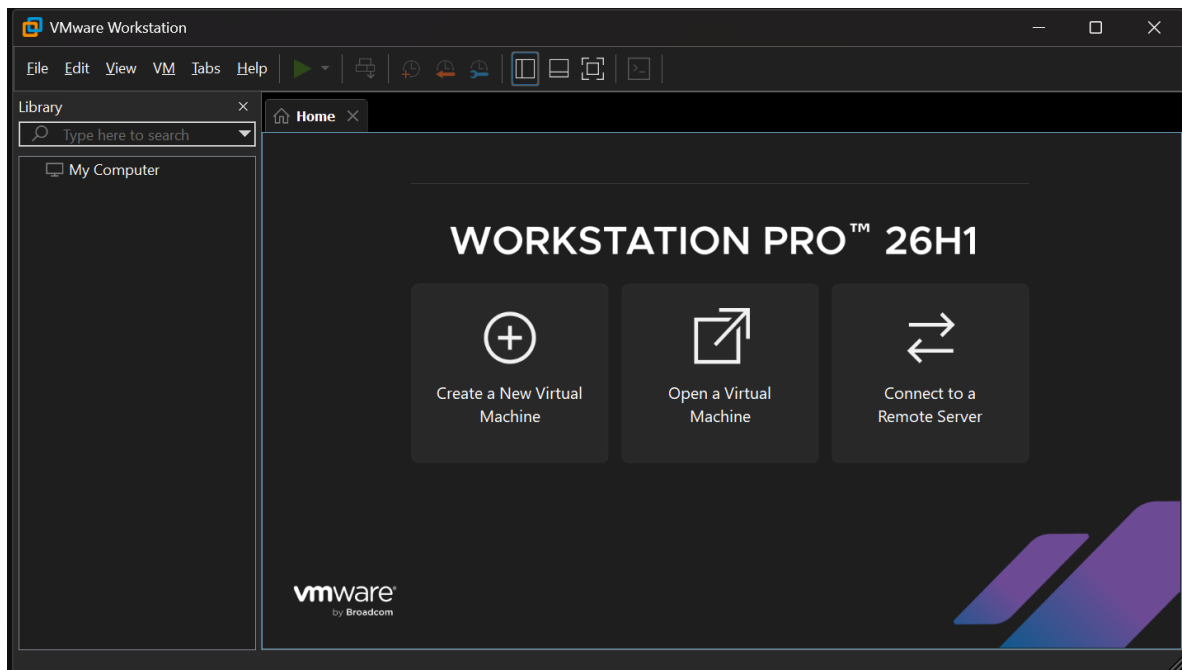
Not published

Cloud Computing (SE-665) Section A

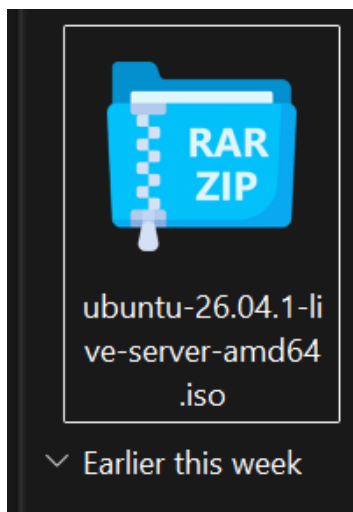
ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Marks have not been published.			
Total	—	—	—

Task 3: Download VMware Workstation Pro and Ubuntu Server

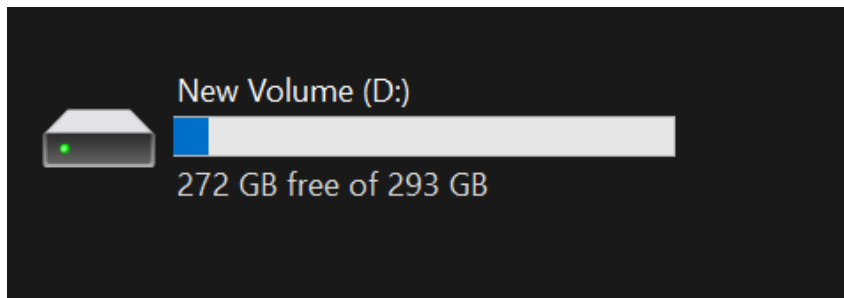
VMware Workstation Pro application running successfully on the Windows host machine.



File explorer window showing the successfully downloaded Ubuntu Server LTS ISO file.



System storage property window verifying more than 20 GB of available disk space for the virtual machine.

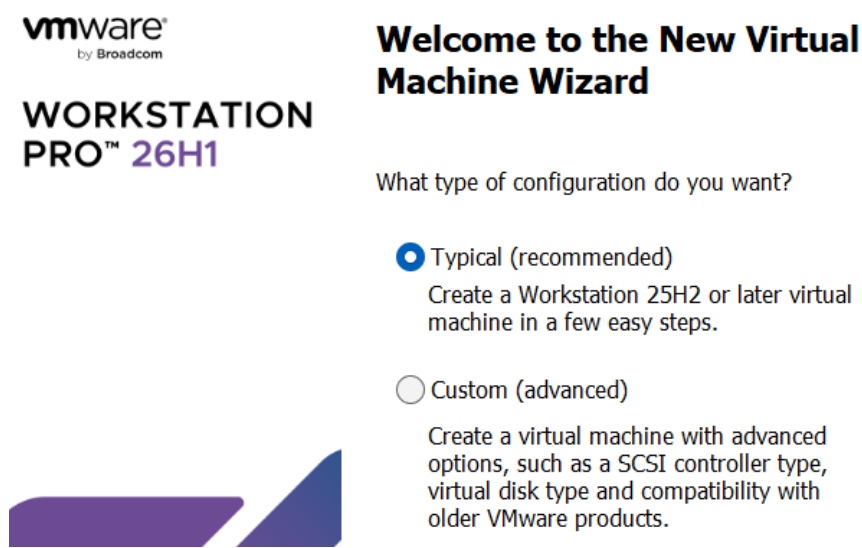


Task 4: Create the Ubuntu Server Virtual Machine

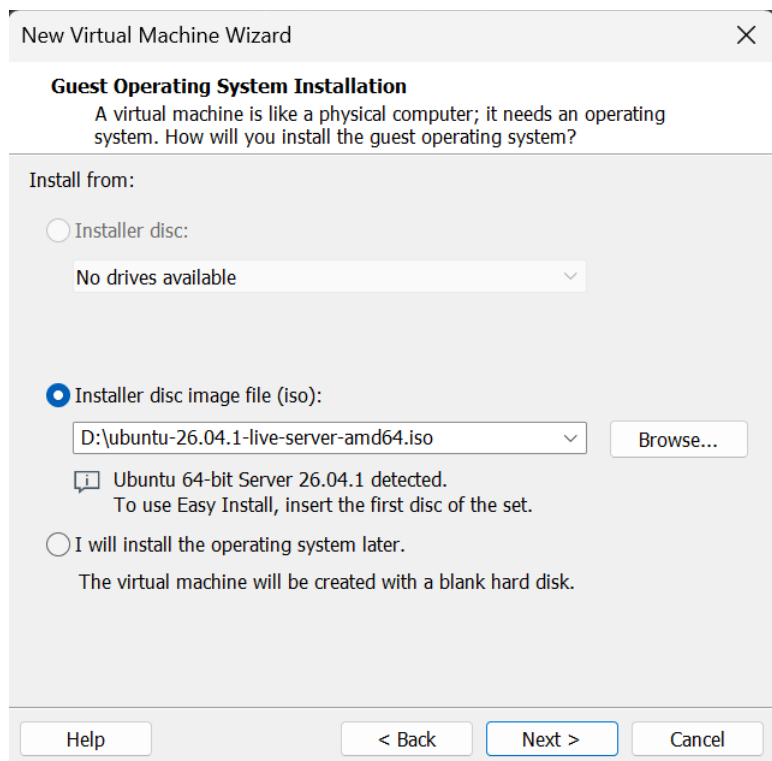
VMware New Virtual Machine Wizard interface open to create a new VM configuration.



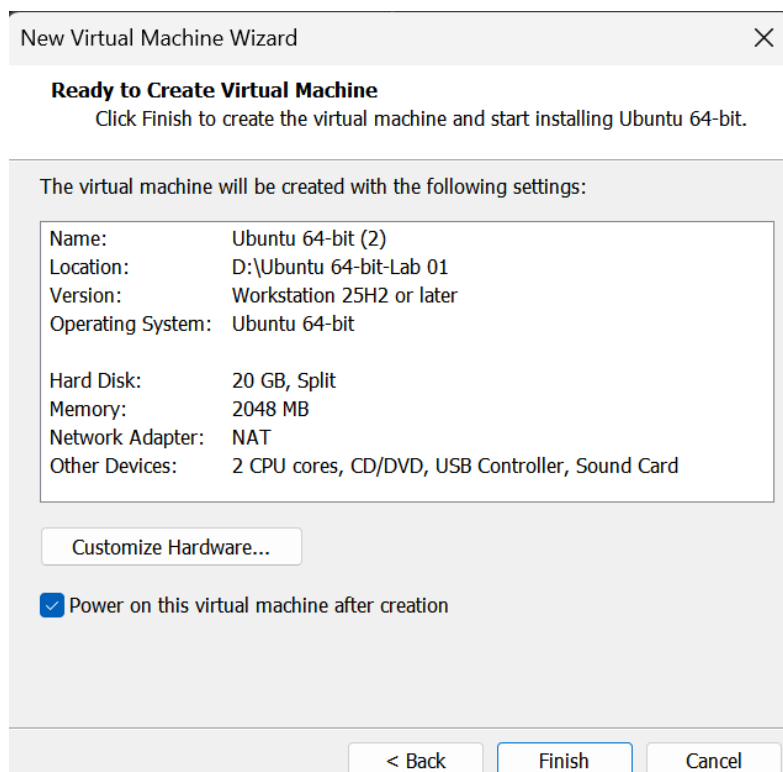
Virtual machine configuration step selecting the recommended Typical setup option.



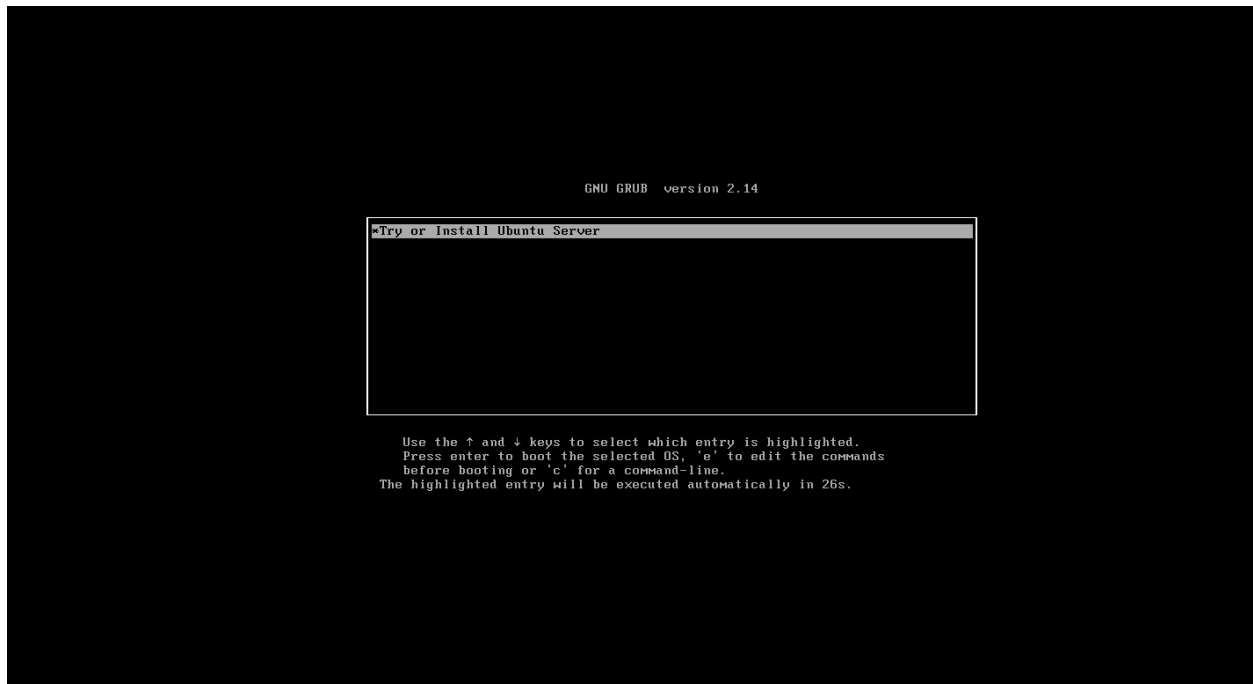
Installer disc image selection step showing the mounted Ubuntu Server ISO path.



Final virtual machine hardware configuration summary allocating at least 20 GB of virtual disk space.

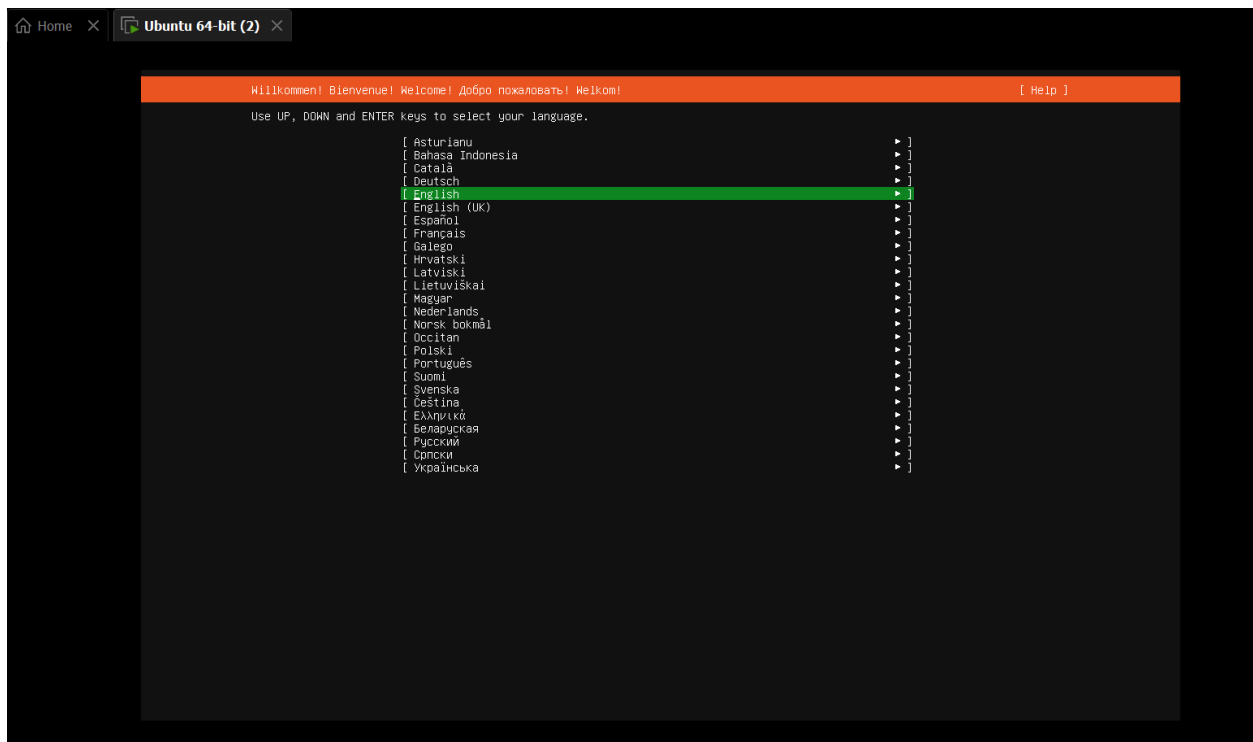


Powered-on virtual machine displaying the Ubuntu Server installer boot screen.



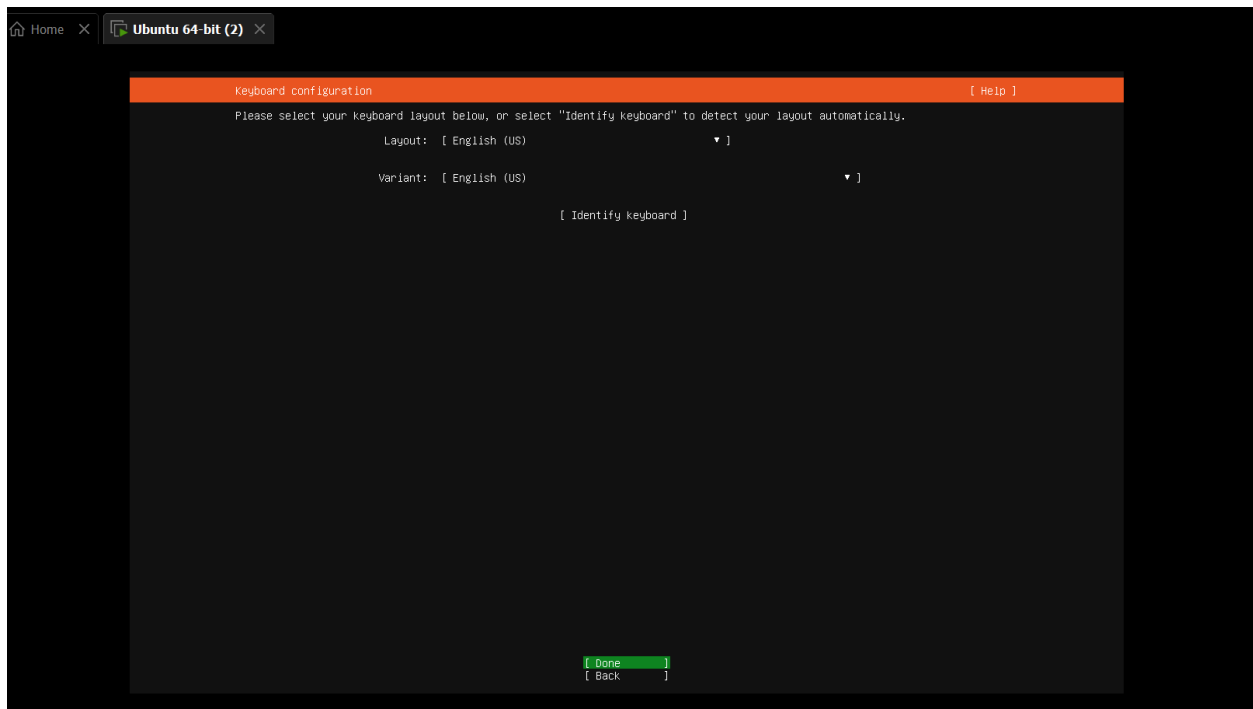
Task 5: Choose the Installation Language

Ubuntu Server installer language selection screen highlighting English as the preferred language.

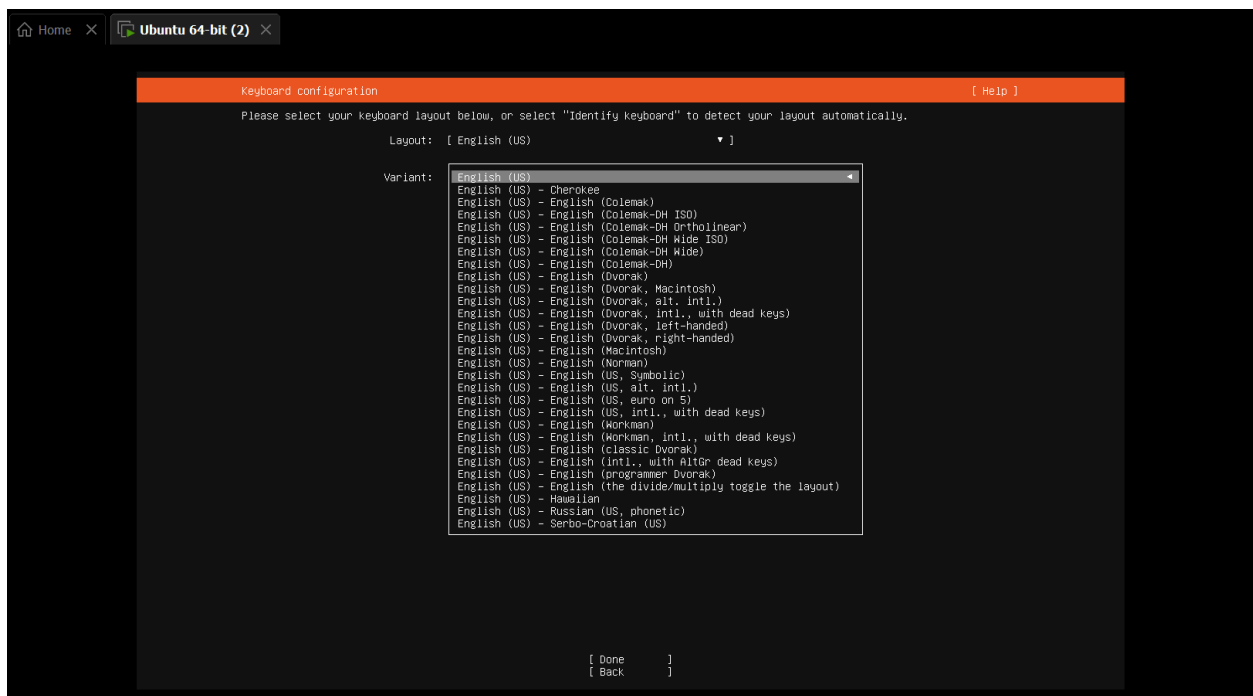


Task 6: Configure the Keyboard Layout

Ubuntu Server keyboard configuration screen showing the selected layout set to English (US).

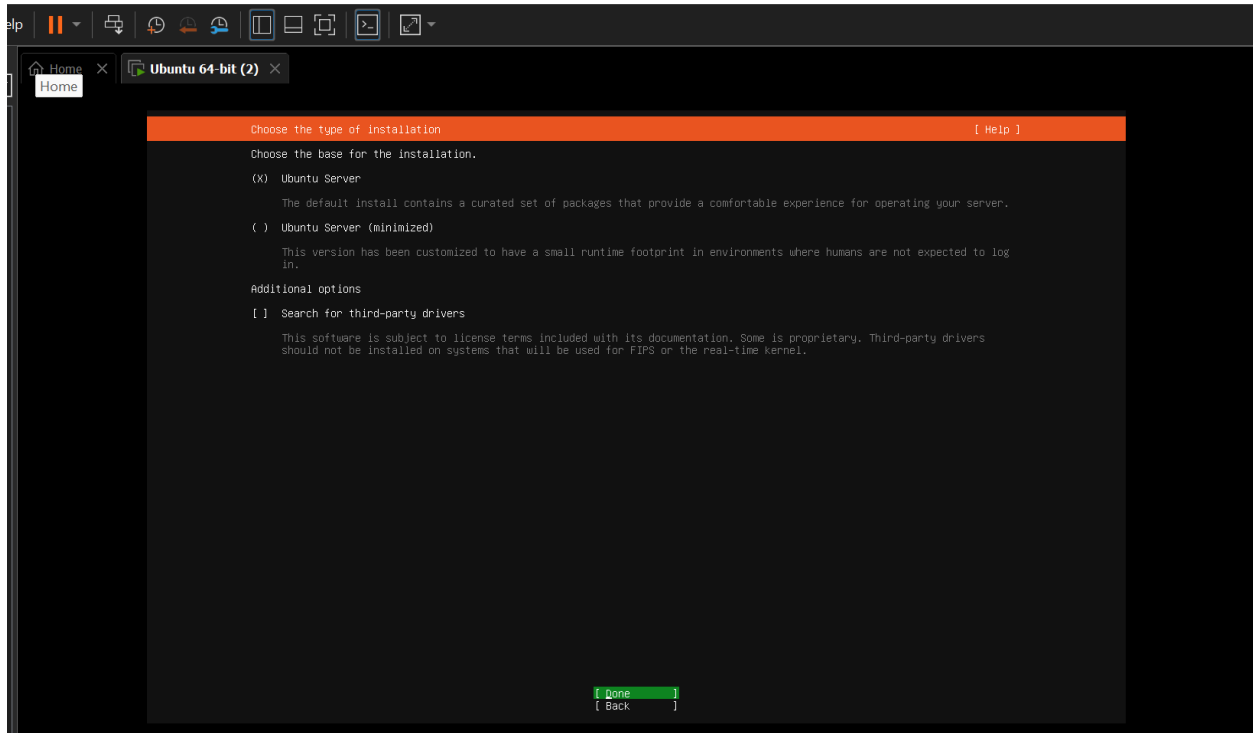


Keyboard configuration options displaying the selected variant configuration.



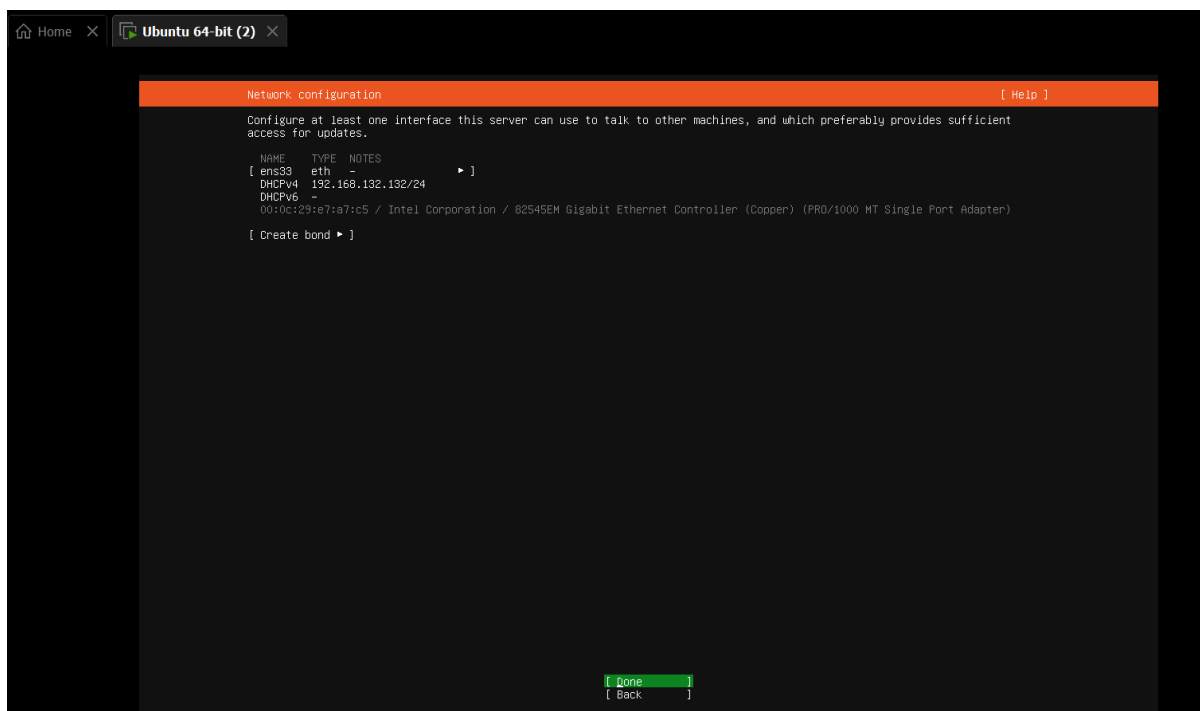
Task 7: Choose the Ubuntu Server Installation Type

Ubuntu Server installation type selection menu highlighting the standard Install Ubuntu option.

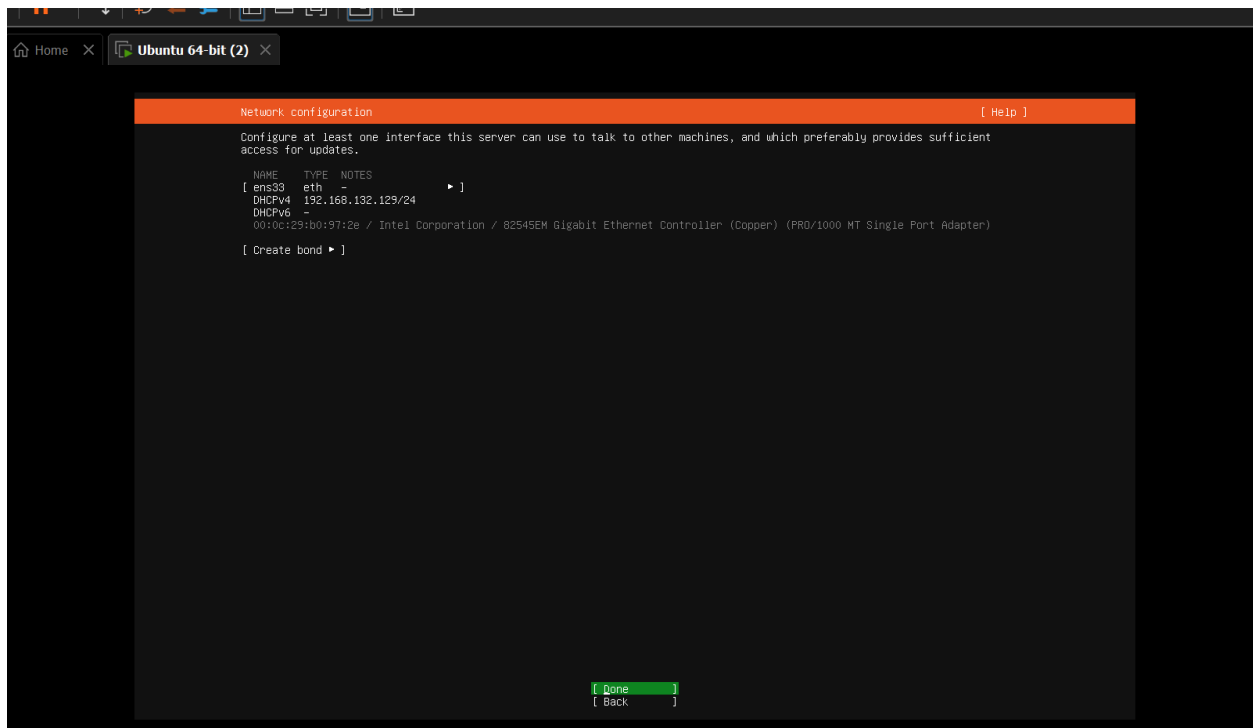


Task 8: Configure the Network

Network connections screen showing the detected ens33 network interface.

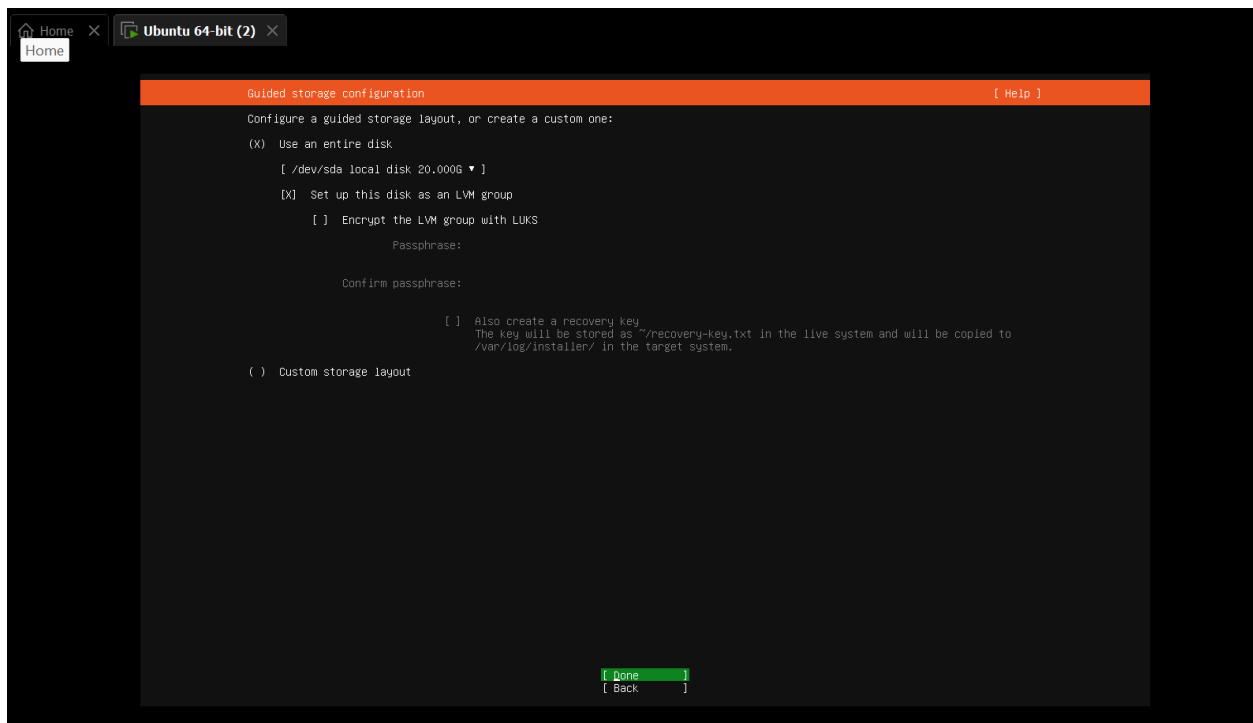


Network configuration details confirming the IPv4 address automatically assigned via DHCP.

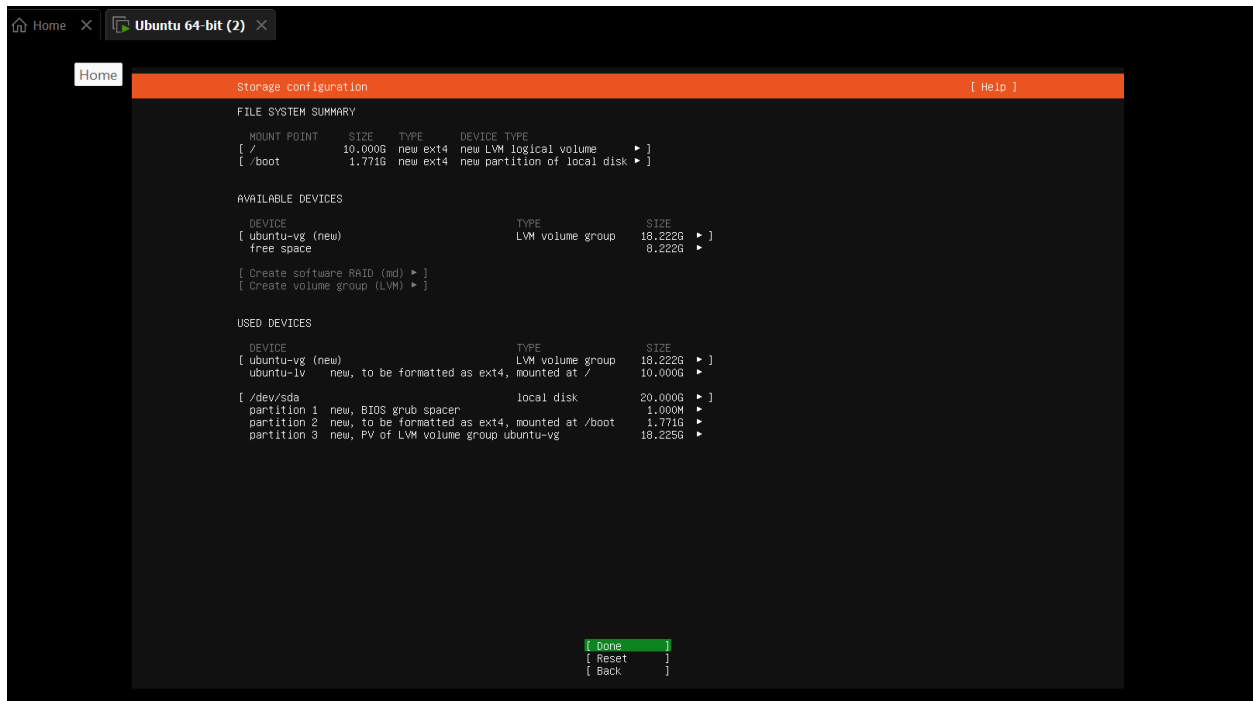


Task 9: Configure Storage

Filesystem setup screen highlighting the guided storage option to use the entire virtual disk.

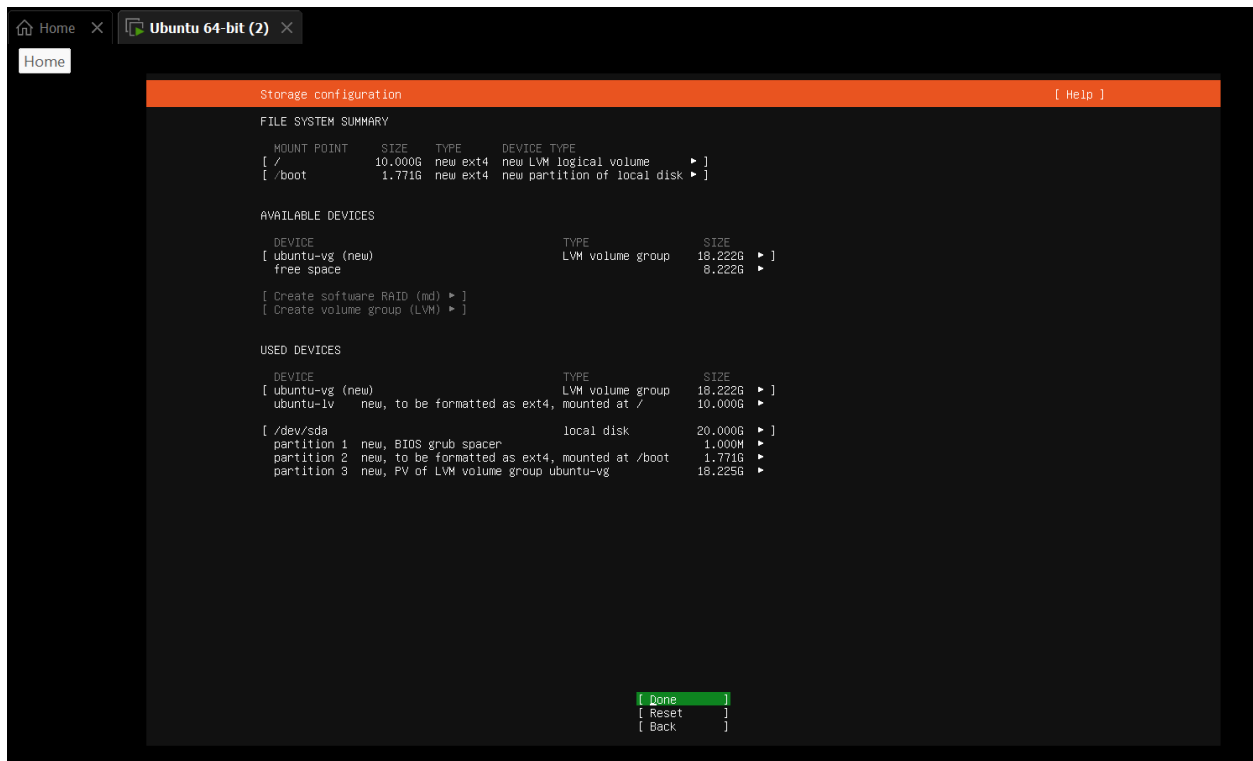


Storage configuration review screen displaying the partition layout before continuing the installation.



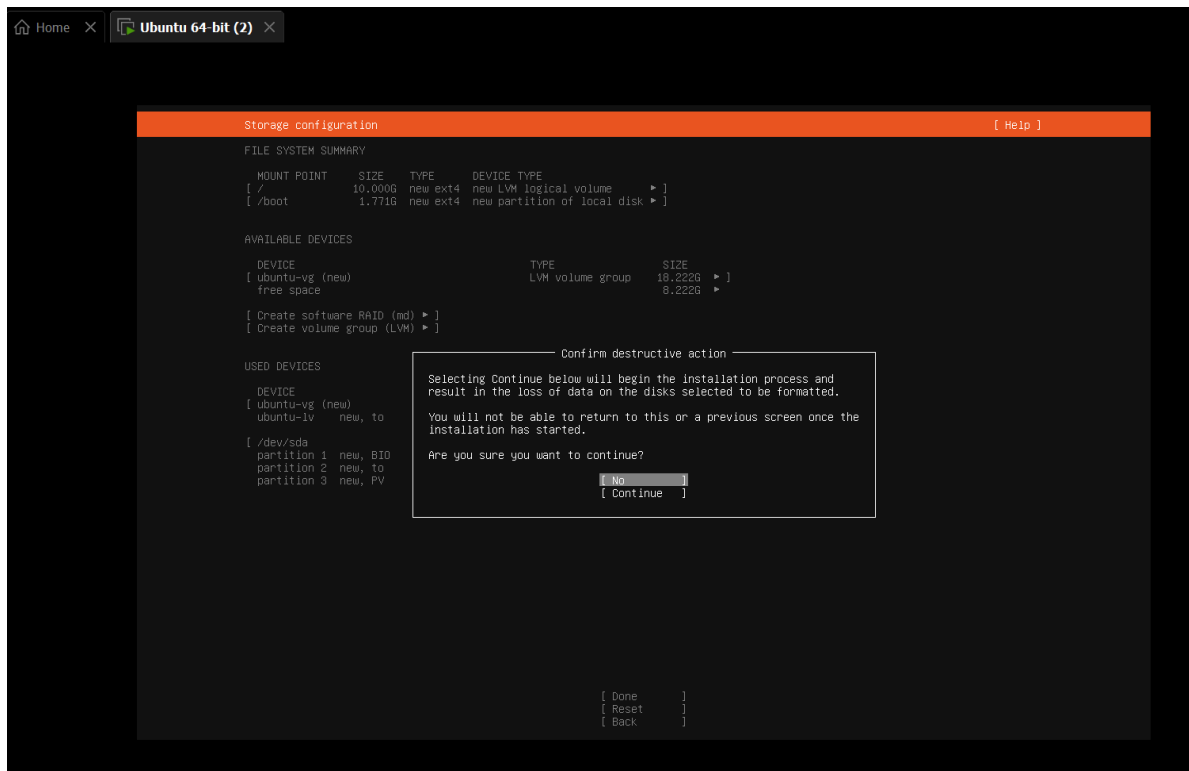
Task 11: Review the Partitions

Filesystem setup summary displaying the proposed virtual disk partitions and mount points.

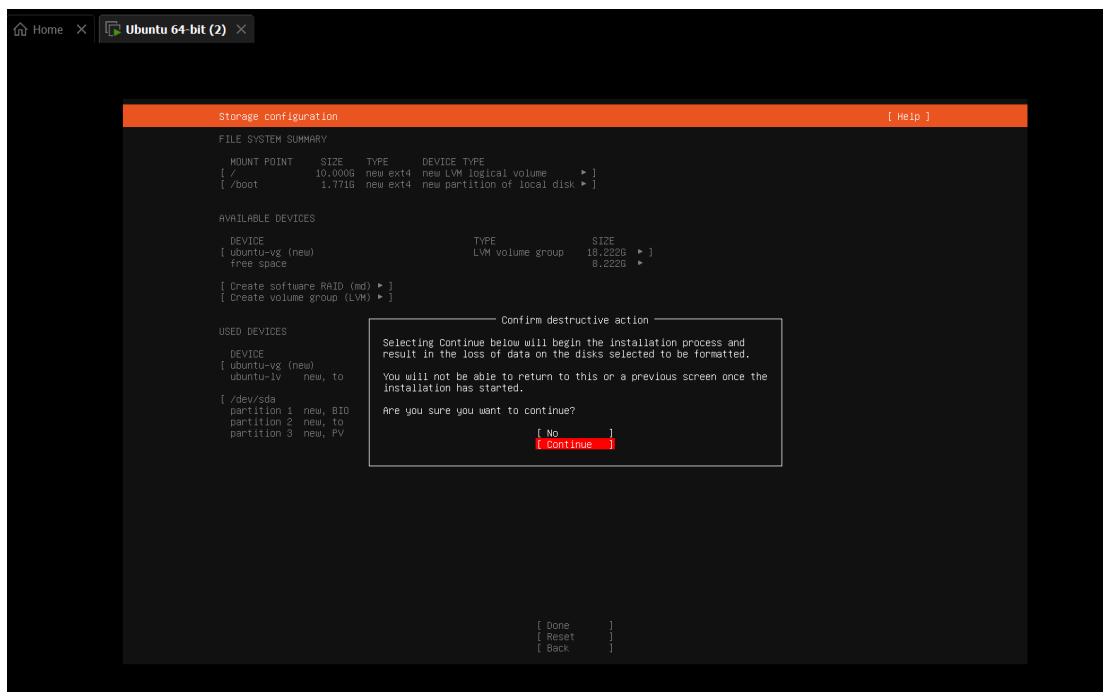


Task 12: Confirm the Storage Changes

Destructive action warning prompt displayed by the installer before formatting the selected disk.

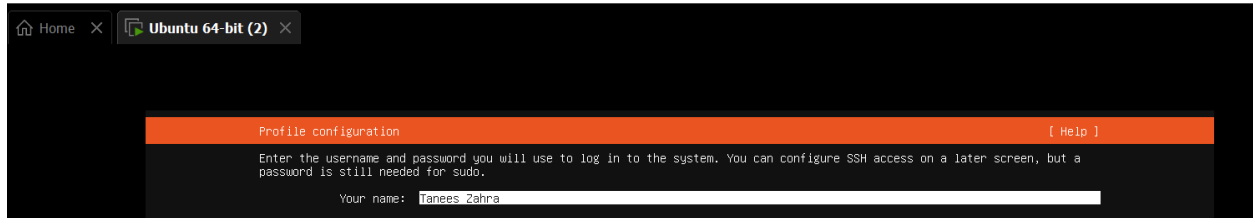


Storage confirmation screen highlighting the selected 'Continue' option to proceed with the installation.

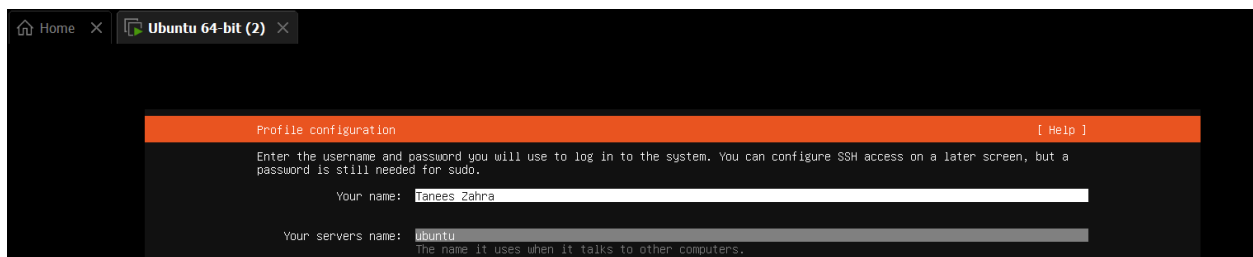


Task 13: Configure the Ubuntu Profile

Ubuntu profile configuration screen showing the user's complete actual name entered in the name field.



Profile setup screen displaying the server name configured as ubuntu.



Profile configuration screen showing the exact GitHub username entered in the username field.



Completed Ubuntu profile setup page showing all configured fields before proceeding to the next step.

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

Task 14: Complete the Software Installation

Ubuntu Server system installation progress showing curtin installation commands and disk configuration logs.

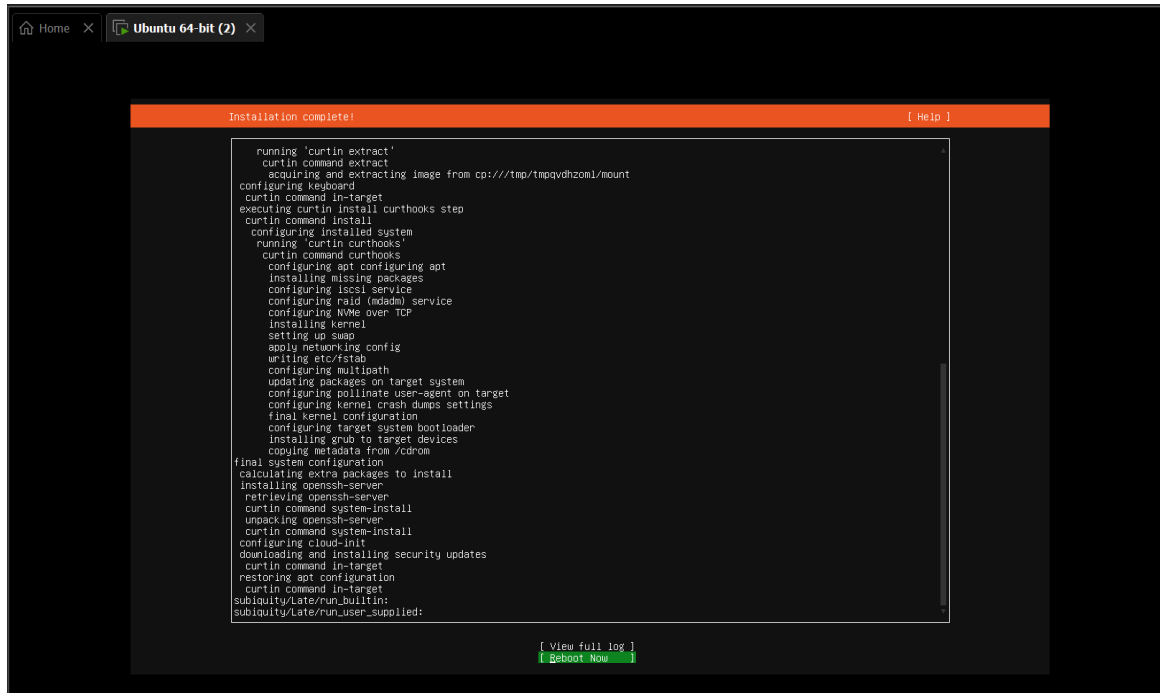
Installing system [Help]

```
subiquity/2dew/apply_autoinstall_config:
subiquity/ad/apply_autoinstall_config:
subiquity/late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
configuring storage
  running 'curtin block-meta simple'
  curtin command block-meta
  removing previous storage devices
  configuring disk: disk-sda
  configuring partition: partition-0
  configuring partitions: partition-1
  configuring partition: partition-2
  configuring format: format-0
  configuring lvm_voigroup: lvm_voigroup-0
  configuring lvm_partition: lvm_partition-0
  configuring format: format-1
  configuring mount: mount-1
  configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  curtin command extract
  acquiring and extracting image from cp:///tmp/tapqvdh2om1/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel -
```

[View full log]

Task 15: Reboot and Verify the Ubuntu Identity

Installation complete screen indicating that the Ubuntu Server setup has successfully finished.

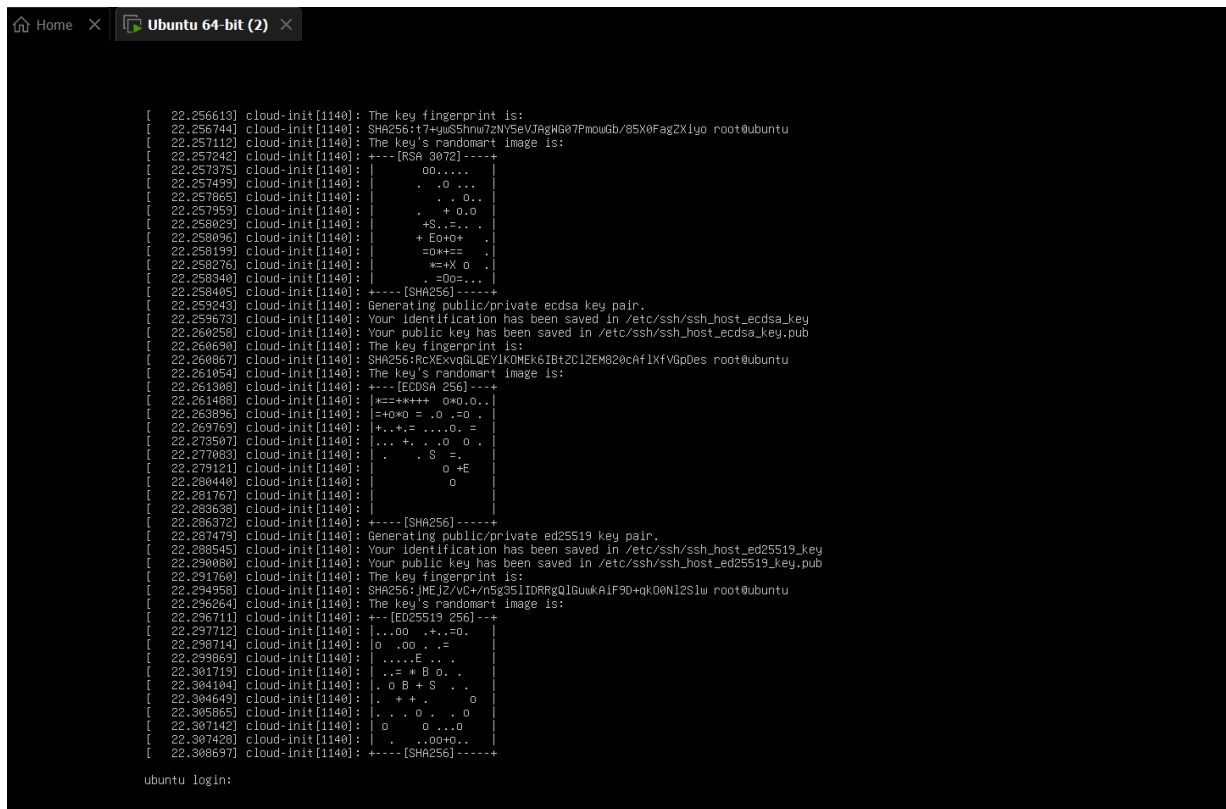


```
Installation complete! [ Help ]

running 'curtin extract'
  curtin command extract
acquiring and extracting image from cp:///tmp/tmpqvdhzm1/mount
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
  configuring apt
  configuring apt
  installing missing packages
  configuring lsscsi service
  configuring raid (mdadm) service
  configuring NVMe over TCP
  installing kernel
  setting up swapt
  apply networking config
  luring etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  configuring kernel crash dumps settings
  final kernel configuration
  configuring target system bootloader
  installing grub to target devices
  copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/run_builtins:
subiquity/late/run_user_supplied:
```

[View full log]
[Reboot Now]

Server login screen displayed after the system restart.



```
[ 22.256613] cloud-init[1140]: The key fingerprint is:
[ 22.256744] cloud-init[1140]: SHA256:t7+ywS5hnu72MY5eVTA9W607PmouGh/85X0Fag2Xl1yo root@ubuntu
[ 22.257112] cloud-init[1140]: The key's randomart image is:
[ 22.257242] cloud-init[1140]: +---[RSA 3072]-----+
[ 22.257375] cloud-init[1140]: |
[ 22.257495] cloud-init[1140]: | .o .o . |
[ 22.257655] cloud-init[1140]: | . . o . |
[ 22.257953] cloud-init[1140]: | + o o |
[ 22.258029] cloud-init[1140]: | +S . . . |
[ 22.258096] cloud-init[1140]: | + Eo+o+ . |
[ 22.258199] cloud-init[1140]: | =o+=+ = |
[ 22.258276] cloud-init[1140]: | *=X o . |
[ 22.258348] cloud-init[1140]: | -o+=... |
[ 22.258405] cloud-init[1140]: +---[SHA256]-----+
[ 22.259243] cloud-init[1140]: Generating public/private ecdsa key pair.
[ 22.259373] cloud-init[1140]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 22.260258] cloud-init[1140]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 22.260630] cloud-init[1140]: The key fingerprint is:
[ 22.260677] cloud-init[1140]: SHA256:RcXExvgGLQEYIKOMEK61Bt2C12EM820cAf1XfV6pDes root@ubuntu
[ 22.261054] cloud-init[1140]: The key's randomart image is:
[ 22.261808] cloud-init[1140]: +---[ECDSA 256]-----+
[ 22.261488] cloud-init[1140]: |*==+++++ O+O.O..|
[ 22.263896] cloud-init[1140]: |+=o+o = .o .o = |
[ 22.269769] cloud-init[1140]: |+.+. = ...o. = |
[ 22.273507] cloud-init[1140]: |...+. .o o . |
[ 22.277083] cloud-init[1140]: | . S = |
[ 22.279121] cloud-init[1140]: | o +E |
[ 22.280448] cloud-init[1140]: | o |
[ 22.281767] cloud-init[1140]: |
[ 22.283638] cloud-init[1140]: +---[SHA256]-----+
[ 22.286372] cloud-init[1140]: Generating public/private ed25519 key pair.
[ 22.287479] cloud-init[1140]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 22.290080] cloud-init[1140]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 22.291760] cloud-init[1140]: The key fingerprint is:
[ 22.294958] cloud-init[1140]: SHA256:jMEJ2/vC+/n5g3S1IDRRgQLGuwKAlF90+qk00N12S1w root@ubuntu
[ 22.296264] cloud-init[1140]: The key's randomart image is:
[ 22.296711] cloud-init[1140]: +---[ED25519 256]---+
[ 22.297712] cloud-init[1140]: |...oo .+. =o. |
[ 22.298714] cloud-init[1140]: |o .oo . = |
[ 22.299863] cloud-init[1140]: |...E . |
[ 22.301719] cloud-init[1140]: |..* B O . |
[ 22.304104] cloud-init[1140]: |o B + S . |
[ 22.304649] cloud-init[1140]: |. + + . o |
[ 22.305865] cloud-init[1140]: |. . o . . o |
[ 22.307142] cloud-init[1140]: |o o ...o |
[ 22.307428] cloud-init[1140]: |. .oo+o.. |
[ 22.308697] cloud-init[1140]: +---[SHA256]-----+

ubuntu login:
```

Successful terminal login prompt showing the authenticated user session.

```
Home x Ubuntu 64-bit (2) x

[ 22.256744] cloud-init[1140]: SHA256:t7+yuSShnu7zNYSeVJagWG07PmowGb/8SX0FagZXIyo root@ubuntu
[ 22.257112] cloud-init[1140]: The key's randomart image is:
[ 22.257242] cloud-init[1140]: +---[RSA 3072]-----+
[ 22.257375] cloud-init[1140]: | oo..... |
[ 22.257499] cloud-init[1140]: | .o.... |
[ 22.257865] cloud-init[1140]: | . .o.. |
[ 22.257959] cloud-init[1140]: | . + o.o |
[ 22.258029] cloud-init[1140]: | +S..+.. |
[ 22.258096] cloud-init[1140]: | +Eo+o+ . |
[ 22.258199] cloud-init[1140]: | =o*+= . |
[ 22.258276] cloud-init[1140]: | *+=X o . |
[ 22.258340] cloud-init[1140]: | =0o=. |
[ 22.258405] cloud-init[1140]: +-----[SHA256]-----+
[ 22.259243] cloud-init[1140]: Generating public/private ecdsa key pair.
[ 22.259673] cloud-init[1140]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 22.260258] cloud-init[1140]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 22.260690] cloud-init[1140]: The key fingerprint is:
[ 22.260877] cloud-init[1140]: SHA256:RcXEvqQLQEY1KOMEK6IBtZC1ZEM820cAfIXfVGpDes root@ubuntu
[ 22.261054] cloud-init[1140]: The key's randomart image is:
[ 22.261308] cloud-init[1140]: +---[ECDSA 256]---+
[ 22.261488] cloud-init[1140]: |*=+*+++ o*o.o.. |
[ 22.263696] cloud-init[1140]: | =+o*o = .o .o . |
[ 22.269769] cloud-init[1140]: | *..+ = ...o. = |
[ 22.273507] cloud-init[1140]: | ...+ . .o o . |
[ 22.277083] cloud-init[1140]: | . . S = . |
[ 22.279121] cloud-init[1140]: | . o +E |
[ 22.280440] cloud-init[1140]: | o |
[ 22.281767] cloud-init[1140]: | |
[ 22.283638] cloud-init[1140]: +-----[SHA256]-----+
[ 22.286372] cloud-init[1140]: Generating public/private ed25519 key pair.
[ 22.288545] cloud-init[1140]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 22.290080] cloud-init[1140]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 22.291760] cloud-init[1140]: The key fingerprint is:
[ 22.294958] cloud-init[1140]: SHA256:JMEJZ/vC+/n5g35IIDRRgQLGuwKAIF90+qk00N12S1w root@ubuntu
[ 22.296264] cloud-init[1140]: The key's randomart image is:
[ 22.296711] cloud-init[1140]: +---[ED25519 256]---+
[ 22.297712] cloud-init[1140]: |...oo .+..=o. |
[ 22.298714] cloud-init[1140]: | o .oo . = |
[ 22.298869] cloud-init[1140]: | ....E . . |
[ 22.301719] cloud-init[1140]: | ..= * B o . |
[ 22.304104] cloud-init[1140]: | . o B + S . |
[ 22.304649] cloud-init[1140]: | . + + . o |
[ 22.305865] cloud-init[1140]: | . . . o . . o |
[ 22.307142] cloud-init[1140]: | o o ...o |
[ 22.307428] cloud-init[1140]: | . .oo+o.. |
[ 22.308697] cloud-init[1140]: +-----[SHA256]-----+

ubuntu login: taneeszahra
Password: _
```

Terminal prompt verifying the correct format displaying the GitHub username and hostname.

```
Home X Ubuntu 64-bit (2) X

[ 22.288545] cloud-init[1140]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 22.290080] cloud-init[1140]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 22.291760] cloud-init[1140]: The key fingerprint is:
[ 22.294958] cloud-init[1140]: SHA256:JMEj2/vC+/n5g351IDRRgQlGuwKAIF9D+qk00NI2Slw root@ubuntu
[ 22.296264] cloud-init[1140]: The key's randomart image is:
[ 22.296711] cloud-init[1140]: +--[ED25519 256]--+
[ 22.297712] cloud-init[1140]: |...oo .+..=o.|
[ 22.298714] cloud-init[1140]: |o .oo . .|=
[ 22.299869] cloud-init[1140]: |.....E...|
[ 22.301719] cloud-init[1140]: |..= * B o. .|
[ 22.304104] cloud-init[1140]: |. o B + S . .|
[ 22.304649] cloud-init[1140]: |. + + . . o|
[ 22.305865] cloud-init[1140]: |. . . o . . o|
[ 22.307142] cloud-init[1140]: |o . o ...o|
[ 22.307428] cloud-init[1140]: |. .oo+o..|
[ 22.308697] cloud-init[1140]: +----[SHA256]-----+

ubuntu login: taneesahra
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Sep 12 03:51:08 PM UTC 2026

System load: 0.31          Processes:             251
Usage of /:  50.5% of 9.75GB Users logged in:          0
Memory usage: 22%          IPv4 address for ens33: 192.168.132.133
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

taneesahra@ubuntu:~$ _
```

Task 16: Find the Ubuntu Server IP Address

Terminal output displaying the execution of the ip addr command to inspect network interfaces.

```
Home × Ubuntu 64-bit (2) ×

[ 22.308697] cloud-init[1140]: +----[SHA256]-----+

ubuntu login: taneeszahra
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Sep 12 03:51:00 PM UTC 2026

System load:  0.31           Processes:      251
Usage of /:   50.5% of 9.75GB Users logged in:    0
Memory usage: 22%          IPv4 address for ens33: 192.168.132.133
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

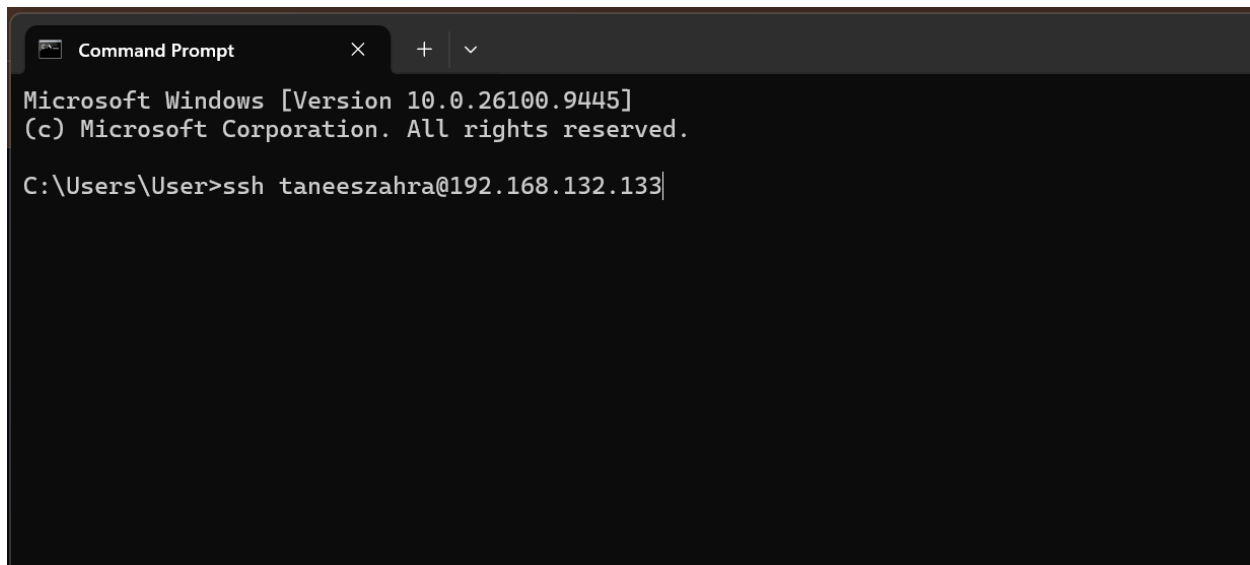
taneeszahra@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:e7:a7:c5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29e7a7c5
    inet 192.168.132.133/24 metric 100 brd 192.168.132.255 scope global dynamic ens33
        valid_lft 1337sec preferred_lft 1337sec
    inet6 fe80::20c:29ff:fe7:a7c5/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
taneeszahra@ubuntu:~$
```

Terminal view highlighting the active IPv4 address assigned to the server's network interface.

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:e7:a7:c5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29e7a7c5
    inet 192.168.132.133/24 metric 100 brd 192.168.132.255 scope global dynamic ens33
        valid_lft 1337sec preferred_lft 1337sec
    inet6 fe80::20c:29ff:fe7:a7c5/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
taneeszahra@ubuntu:~$
```

Task 17: Connect to Ubuntu Server from Windows

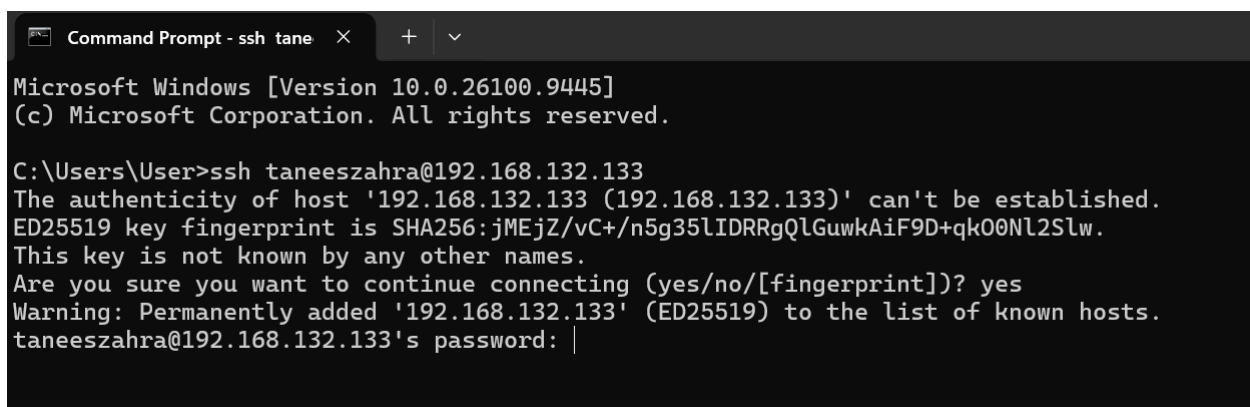
Windows Command Prompt showing the SSH connection command entered using the remote server's IP address.



```
Command Prompt
Microsoft Windows [Version 10.0.26100.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh taneesahra@192.168.132.133
```

Terminal session displaying the prompt to accept the remote server's authenticity fingerprint.



```
Command Prompt - ssh tane
Microsoft Windows [Version 10.0.26100.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh taneesahra@192.168.132.133
The authenticity of host '192.168.132.133 (192.168.132.133)' can't be established.
ED25519 key fingerprint is SHA256:jMEjZ/vC+/n5g35lIDRRgQlGuwkAiF9D+qk00Nl2Slw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.132.133' (ED25519) to the list of known hosts.
taneesahra@192.168.132.133's password: |
```

Successful remote SSH session established from the Windows host to the Ubuntu Server.

```
taneeszahra@ubuntu: ~  
Warning: Permanently added '192.168.132.133' (ED25519) to the list of known hosts.  
taneeszahra@192.168.132.133's password:  
Connection closed by 192.168.132.133 port 22  
  
C:\Users\User>ssh taneeszahra@192.168.132.133  
taneeszahra@192.168.132.133's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Sat Sep 12 04:11:01 PM UTC 2026  
  
System load:  0.14          Processes:            224  
Usage of /:   50.8% of 9.75GB Users logged in:          0  
Memory usage: 31%          IPv4 address for ens33: 192.168.132.133  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
8 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
taneeszahra@ubuntu:~$ |
```

Remote terminal prompt verifying the correct authenticated user identity and server hostname.

```
taneeszahra@ubuntu: ~  
Expanded Security Maintenance for Applications is not enabled.  
  
8 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
taneeszahra@ubuntu:~$ |
```